SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305
Name:T Gnaneswari
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COURSE OUTCOME COVERED

CO 4: Formulate data-driven web solutions using XML, XSLT, and JSP within the MVC framework. (L4)

Assessment Tool: Class Test

Weightage: 10%

QUESTIONS: **Total Marks: 20**

1. Explain the fundamental concepts used in web development by defining the following terms with suitable explanations: **XML, JSP, MVC architecture, XSLT, XML Schema, JSTL, PHP, Database Connectivity, and Parser**. Your answer should clearly describe the purpose and role of each concept in web applications.

2. Explain how the above technologies are interconnected in building a web application. Describe how XML is structured and validated, how JSP and PHP are used for server-side processing, how MVC architecture organizes the application, and how database connectivity and parsers contribute to data processing and communication.

Code of Conduct:

I T Gnaneswari(192372183) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

T Gnaneswari

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	25%	Clearly explains all concepts with complete accuracy and depth	Minor gaps in understanding	Basic understanding with some errors	Incorrect or very limited understanding
Content Coverage	25%	Covers all required topics comprehensively	Covers most topics with minor omissions	Covers some topics only	Very few topics covered
Technical Accuracy	20%	All explanations are technically correct and precise	Minor technical errors	Some incorrect or vague explanations	Major technical errors
Integration of Concepts	15%	Effectively connects and relates concepts logically	Some connections made with minor gaps	Limited connections between concepts	No clear relationship between concepts
Clarity & Presentation	15%	Well-structured, clear, and easy to understand	Mostly clear with minor issues	Somewhat unclear or poorly organized	Poorly structured and difficult to understand

22/08/26

CO4 - AT3 - Class Test
Internet programming

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1) fundamental concepts used in web development

web applications use different technologies to store data, process user requests, communicate with databases, and display dynamic information. Important concepts include XML, JSP, MVC, XSLT, XML schema, JSTL, PHP, Database connectively, and parser.

1. XML - Extensible Markup language

XML (Extensible Markup language) is a markup language used to store, organize and exchange structed data. Unlike html, XML does not have predifined tags. Developers can create their own tags according to the data being represented.

Example:

```
<Student>
    <name> Gnaneswari </name>
    <regno> 192372183 </regno>
    <dept> CSE-AI </dept>
</student>
```

<Student> is root element.

<name>, <regno>, <dept> are children element.

Purpose in web application:

* Stores structred information.

2. Transfer data b/w different appications.
3. provides a platform - independent data format.
4. can be validated using XML schema.
5. can be transformed using XSLT.
6. Can be read using XML parsers.

2. JSP - Java server pages

JSP (Java Server page) is a Java-based server-side technology used to create dynamic web pages. A JSP used page can combine HTML with Java-based server-side processing.

Example:

```
<html>
 <body>
  <h2>welcome</h2>
  <%
     string name = request.getParameter ("name");
     out.println (" Hello" + name);
  %>
</body>
</html>
```

Purpose:

* Generate dynamic HTML pages.
* process user requests
* Reterives form value.
* work with servelets and Jave Beans.

3. MVC Architecture

MVC stands for Model view controller. It is an architectural pattern used to divide a web application into three major components.

model

Handles data, database, operations, logic.

view

Dispay information to the user.

controller

Receives requests and coordinates model and view.

User
↓
Controller
↓
model
↓
Database
↓
Model
↓
Controller
↓
View
↓
User

4. XSLT - Extensible stylesheet Language Transformations.

XSLT is a language used to transform XML documents into another format such as XML, HTML.

XML data.
↓
XSLT stylesheet
↓
XSLT processor
↓
HTML output

Purpose :-

* Transforms XML into HTML
* changes one XML structure into another.

5. XML schema - XSD

XML schema, commonly called XSD (XML schema Definition) define structure and rules that an XML document must follow.

- Allowed element
- Allowed attributes.
- Data types
- element order.
- optional elements
- Number of occurence.

example:

<xs:element name ="age" type= "xs:integer"/>

XML Document

↓

XML schema

↓

validation

↓

valid/Invaild XML

6. JSTL - JSP Standard Tag Library

JSP (JSP standard Tag library) is a conection of Standard tags used in JSP pages.

It reduce need to write Java code directly.

Example:

```
<c:foreach var="student" items="${Students}">
    ${student.name}
</c:forEach>
```

JSTL provides Tags for

- * conditions
- * Loops
- * URL handling.
- * XML processing
- * formatting
- * Internationalization.

7) PHP - Hypertext preprocessor

PHP is a server-side scripting process language used to develop dynamic web applications.

Example

```
<? PHP
  $name = $_POST ["name"];
  echo "welcome". $name;
?>
```

Purpose:

Process HTML forms.

Handle user authentication.

Manage sessions and cookies.

Perform calculations.

8) Database connectivity

Database connectivity refers to estabishing communication between a web application and a database.

Examples of database:

MySQL

PostgreSQL

Oracle

SQL Server

Application
↓
Database Driver
↓
Database connection
↓
SQL Query
↓
Database
↓
Result

main operations

C - create.

R - Read

U - update.

D - Delete.

9. parser

A parser is a softwaer component that reads Structrued data and converts into a from that an appiication can understand and process.

DOM parser

DOM (Document object model) loads complete XML document

```
Student
   ├ name
   ├ reg no
   └ Department.
```

SAX parser

SAX (simple API for XML) process XML sequatially using events without loading the compiete document into memory.

Purpose:

Read XML documents.

Access element and attributes.

Extract values.

2) Interconnection of these Technologies in a Web Application.

These technologies can work together to create a compiete web appiication. Each technology pattern performs a different responsibility.

Overall Architecture:

User/web browser

↓

controller

servlet / PHP Logic.

MODEL

Business logic

↓

DATABASE

MYSQL /oracle.

XML processing

parser + XSD.

↓

XML Document.

↓

XSLT.

↓

view

JSP + JSTL

or ~~PHP~~ PHP/HTML

↓

HTML Response.

↓

USER.

Step 1:

XML store structured data- information.

<Product>

<id> 101</id>

<name> rajesh</name>

<age> 45 </age>.

</product>.

Step 2:

XML schema validates the XML.

Example:

Product ID → Integer
Name → String
Price → Decimal.

<Price> ABC </Price>

XML
↓
XML schema validation.
↓
Valid ?
↓
YES → continue processing
NO → Display Error.

Step 3:

Parser Reads XML.

Once XML data is valid, an XML parser can read it.

XML file
↓
DOM / SAX parser
↓
Extract Elements
↓
Application Data

ID = 101
Name = rajesh
age = 45

Step 4:

XSLT transform XML

If XML Transform needs to be displayed directly, XSLT can transform it into HTML.

XML

+

XSLT

↓

Transformation

↓

HTML

↓

Browser

Step 5:

JSP or PHP performs server-side programming

JSP-Based application.

Browser → servelt/controller → Model → Database

↓

JSP+JSTL

↓

Browser

PHP-Based application

Browser

↓

PHP Script.

↓

Business Logic

↓

Database

↓

PHP Generate HTML

↓

Browser

Step 6: MVC organises the Application

model	view	controller.
logic	pages	Browsing requests.

Step 7: Database connectivity.

Example: Student Result web Application

Register no: 192372183

⇓

Controller Server request

⇓

Model process request.

⇓

JDBC connects to database

⇓

Students result retrieved

⇓

XML may requests / Exchange result

⇓

XSD validates XML structure.

⇓

parser reads XML

⇓

XSLT may transform XML into HTML

⇓

JSP + JSTL display the result.

⇓

Students sees result in browser